PLamb Cosmology Analysis – Master Summary Report

This document consolidates all recorded runs from the PLamb Test V10V6c_plus framework. It integrates previously sorted BIC tables with additional unsorted results, producing a master view of model comparisons.

The focus is on Supernovae (Pantheon+SH0ES), BAO, and Planck distance priors, across standard (ΛCDM, KdS) and extended (FR, PBH) models.

Variables and Parameters

Variable	Description
H0	Hubble constant today [km/s/Mpc]
Ωm (Om)	Matter density parameter
ΩΛ (Ol)	Dark energy (cosmological constant) density parameter
Ωk (Ok)	Spatial curvature parameter, 1 - Ωm - ΩΛ
A_acc	Effective acceleration fluid amplitude (KdS/FR extension)
n_acc	Acceleration fluid exponent (power-law index)
γ_c	Variable-c parameter (redshift dependence of speed of light)
ε_M	Magnitude evolution parameter (variable-M of SN absolute mag)
ΔM(z)	Luminosity evolution law, linear_z or log1pz form
χ^2	Chi-squared statistic
AIC	Akaike Information Criterion (model comparison)
BIC	Bayesian Information Criterion (model

	comparison)
dof	Degrees of freedom (N_data - N_params)
w_eff	Effective equation-of-state proxy derived from n_acc

Acronyms

Acronym	Meaning
SN	Type Ia Supernovae
BAO	Baryon Acoustic Oscillations
CMB	Cosmic Microwave Background
CC	Cosmic Chronometers (H(z) measurements)
FR	Full Relativity framework (generalized Kerr–de Sitter)
KdS	Kerr–de Sitter cosmology (rotating black hole extension)
PBH	Parent Black Hole accretion-driven cosmology (exploratory model)
LCDM	Standard cosmological model with Λ and Cold Dark Matter
r_d	Sound horizon at drag epoch
l_A	CMB acoustic angular scale
R	CMB shift parameter

Overview of Primary Models

The following model families were tested and compared against standard ΛCDM baselines:

• ΛCDM (Flat & Curved): The standard cosmological model, tested with and without BAO and Planck priors.

• KdS (Kerr–de Sitter): Incorporates an effective acceleration term motivated by black-hole physics.

• FR (Full Relativity): Extends KdS with variable-c (γ_c) and magnitude evolution (ε_M).

• PBH (Parent Black Hole): Exploratory wrapper testing effects of parent black hole accretion on space-time fabric (STF).

Current Status of Runs

Recent results include FR varM (ε_M free) runs with and without Λ, flat/curved tests, and robustness checks with z-cuts.

Preliminary PBH test scripts have been defined but require execution.

Planck distance-prior integration has been set up but requires JSON input (means and covariance).

Model Performance Summary

• Flat ΛCDM with SN+BAO remains a baseline reference (lowest ΔAIC/ΔBIC).

• Curved ΛCDM shows mild preference in some runs but penalized by BIC.

• FR varM(log1pz) shows competitive χ^2 fits with additional freedom, but requires more evidence for parameter necessity.

• KdS models show lower H0 and tension with BAO unless constrained.

• PBH exploratory runs are pending — expected to test accretion-only, propagation-only, and dimension/duality-only effects.